

A Finite Magma Challenge for Equation 677

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Problem 1 (Main question). Let M be a finite nonempty set with a binary operation $\diamond$. Suppose that, for all $x, y \in M$,

$$\text{E677 :} \quad x = y \diamond (x \diamond ((y \diamond x) \diamond y)).$$

Prove that, for every $x \in M$,

$$\text{E255 :} \quad x = ((x \diamond x) \diamond x) \diamond x.$$

The following hints are meant to be used in order. They are deliberately phrased as a proof path, with the target of proving the identity E255 from E677 and finiteness.

Warning. Do not assume associativity, commutativity, an identity element, inverses, right cancellation, or quasigroup structure. The only cancellation allowed in the hints below is left cancellation after it has been proved.

Hints

- 1. First obtain left division.** Fix $a \in M$ and let $L_a(t) = a \diamond t$. From E677, with $y = a$, show that for every $b \in M$,

$$b = a \diamond (b \diamond ((a \diamond b) \diamond a)).$$

Thus L_a is surjective, hence bijective because M is finite. Define $a \backslash b$ to be the unique element u with $a \diamond u = b$. Then

$$a \backslash b = b \diamond ((a \diamond b) \diamond a).$$

All later cancellations should be read as cancellations by one of these bijective left translations.

- 2. Derive the two basic identities.** Show that, for all $y, z \in M$,

$$z = (y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y), \tag{T}$$

and

$$(y \diamond (y \diamond z)) \diamond y = z \diamond (((y \diamond z) \diamond z) \diamond (y \diamond z)). \tag{K}$$

For (T), substitute $y \diamond z$ for x in E677 and cancel y . For (K), compare (T) with E677 after replacing y by $y \diamond z$, then cancel $y \diamond z$.

3. Reduce the problem at a fixed point. Fix $x \in M$ and put

$$a = x \diamond x, \quad q = a \diamond x = ((x \diamond x) \diamond x), \quad p = x \backslash x.$$

The desired identity at x is simply

$$q \diamond x = x. \tag{1}$$

Prove the useful strengthening: if $u \diamond x = x$, then $u = q$. Indeed, (T) gives $x = x \diamond (x \diamond u)$, while E677 at $(x, y) = (x, x)$ gives $x = x \diamond (x \diamond q)$; two left cancellations give $u = q$.

Thus E255 at x is equivalent to the existence of a right fixed point over x . It is also equivalent to each of

$$q \backslash x = x, \quad ((x \diamond x) \diamond q) \diamond (x \diamond x) = x, \quad (q \diamond x) \diamond q = p. \tag{2}$$

Use (T) with $(y, z) = (x \diamond x, x)$, and the formula for $q \backslash x$, to verify these equivalences.

4. Introduce the fixed maps. For the same fixed x , define

$$H(t) = (x \diamond t) \diamond x, \quad F(t) = x \diamond (t \diamond x).$$

Prove

$$x \backslash t = t \diamond H(t), \quad F(L_x(t)) = L_x(H(t)). \tag{3}$$

Hence F and H are conjugate self-maps of M . Show that (1) is equivalent to either of these maps having a fixed point. If a fixed point exists, it is unique.

5. Study the orbit of x under L_x . Let

$$c_0 = x, \quad c_{k+1} = x \diamond c_k.$$

Since L_x is a permutation of the finite set M , the sequence is periodic. Let d be the least positive period of the orbit of x , and read all subscripts modulo d . Put

$$d_k = c_k \diamond x.$$

Using E677 with parameter x , prove the recurrence

$$c_{k-1} = c_k \diamond d_{k+1}. \tag{4}$$

Then show

$$p = c_{d-1}, \quad q = d_1 = c_{d-2}, \quad p \diamond c_1 = q. \tag{5}$$

A further useful fact is this: if $d_j = x$, then $j \equiv d - 2 \pmod{d}$.

6. Eliminate the smallest periods. If $d = 1$, then $x \diamond x = x$, so $q = x$ and (1) follows. The case $d = 2$ contradicts the last statement in Hint 5.

For $d = 3$, write

$$c_0 = x, \quad c_1 = a = q, \quad c_2 = p.$$

Use (4) and (K) to prove successively

$$x = a \diamond (p \diamond x), \quad a \diamond p = x, \quad p \diamond x = p,$$

then

$$(a \diamond a) \diamond a = p \quad \text{and} \quad (a \diamond a) \diamond a = x \diamond (a \diamond a).$$

Cancel on the left by x at the final step to get $a \diamond a = a$, contradicting the displayed identity $(a \diamond a) \diamond a = p$. Thus any unresolved point has orbit period $d \geq 4$.

- 7. Use finiteness on the right.** Assume, for the sake of contradiction, that $q \diamond x \neq x$. By Hint 3, no element $u \in M$ satisfies $u \diamond x = x$. Hence x is missing from the image of the map $R_x(t) = t \diamond x$. Since M is finite, R_x has a nontrivial fiber: there exist $y \neq z$ and $e \in M$ such that

$$y \diamond x = z \diamond x = e. \tag{6}$$

Apply (T) twice, once with $(y, z) = (y, x)$ and once with $(y, z) = (z, x)$. After left cancelling e , obtain

$$(y \diamond e) \diamond y = (z \diamond e) \diamond z = e \setminus x. \tag{7}$$

Moreover, if $y \diamond e = z \diamond e$, then a further left cancellation gives $y = z$. Therefore the map $t \mapsto t \diamond e$ is injective on every such collision fiber.

- 8. Convert a collision into the fixed point.** The clean finishing lemma would be:

$$\boxed{y \diamond x = z \diamond x = e, \quad y \neq z \implies e \diamond x = x.} \tag{C}$$

If (C) is proved, then the problem is solved. Indeed, (6) gives $e \diamond x = x$, and Hint 3 then forces $e = q$, so $q \diamond x = x$.

Here is a route toward (C). Define

$$\Phi(u, v) = (u \diamond v, (u \diamond (u \diamond v)) \diamond u).$$

If $\Phi(u, v) = (r, s)$, prove from (T) that

$$r \diamond s = v. \tag{8}$$

Thus a forward Φ -orbit may be written as

$$\Phi^n(u, v) = (\alpha_n, \alpha_{n+2}), \quad \alpha_n \diamond \alpha_{n+2} = \alpha_{n+1}. \tag{9}$$

For a collision (6), equation (7) gives a common element

$$t = (y \diamond e) \diamond y = (z \diamond e) \diamond z, \quad e \diamond t = x,$$

and hence

$$\Phi(y, x) = \Phi(z, x) = (e, t). \tag{10}$$

It remains to prove $t = x$. Since M is finite, the forward Φ -orbit of (e, t) eventually repeats. The intended last move is to pull the first repeated pair backward through the recurrence (9) until it reaches the two distinct predecessors (y, x) and (z, x) . If this pullback can be justified using only left cancellation, then $t = x$; consequently $e \diamond x = x$, and the main identity follows.

9. Optional orbit test for period four. As a smaller arena for the preceding ideas, suppose the L_x -orbit of x has period 4. Write

$$c_0 = x, \quad c_1 = a, \quad c_2 = q, \quad c_3 = p, \quad r = q \diamond x, \quad s = p \diamond x.$$

Use the recurrence (4), together with (T) and (K), to prove

$$x \diamond a = q, \quad a \diamond x = q, \quad a \diamond r = x, \quad x \diamond q = p, \quad x \diamond p = x, \quad p \diamond a = q,$$

and

$$r \notin \{x, a, q\}. \tag{11}$$

If $r = p$, show further that

$$q \diamond a = q, \quad s = (q \diamond q) \diamond q = a \diamond (q \diamond q), \quad s \notin \{x, a, q, p\}. \tag{12}$$

These identities make period four a good test case for the collision lemma (C): the right tail is forced to leave the principal orbit unless (1) already holds.

Final goal. Prove the boxed collision lemma (C), or prove any one of the equivalent identities in (1)–(2), using the finite repetitions supplied above. Once $q \diamond x = x$ is obtained for the fixed but arbitrary element x , the identity E255 follows for all elements of M .